Cross-Network Intrusion Analysis

Phase 5 — Feature Engineering of the merged NetFlow data

Ahmed Hamouda

Notebook `08_features_netflow.ipynb`

1 Goal and method

Requirement. Explain feature selection, extraction / creation, encoding, and scaling / transformation when appropriate, and justify the final feature set.

Data. The merged NetFlow table of phase 3: 17,551,848 behaviours from four networks (UNSW-NB15, CSE-CIC-IDS2018, ToN-IoT, BoT-IoT). Every choice below is based on a decision of phase 2 or a finding of phase 4.

Steps. (0) starting columns; (1) creation of new features; (2) encoding of the code columns; (3) log transformation and robust scaling; (4) feature selection; (5) final feature set; (6) PCA; (7) techniques not used.

No train/test split. The project stops before modelling. The steps that learn a value from the data (scaling, the list of frequent application protocols, the selection) are fitted on all the data. *In a modelling phase, these steps must be refitted on the training part only, to avoid data leakage.*

Sample. Correlation, mutual information and PCA use one sample of up to 5,000 random behaviours from each network × family group (102,058 rows), so every family counts. All other steps use all 17.5 million rows.

2 Starting columns

Nine columns are left out because they must never be features (phase 2 decisions):

Table 1: Columns left out before feature engineering.

Columns	Reason
`FLOW_START_MILLISECONDS`, `FLOW_END_MILLISECONDS`	reveal the lab's attack schedule (M2)
`L4_SRC_PORT`, `DNS_QUERY_ID`	random numbers chosen by the machine (M5)
`MIN_TTL`, `MAX_TTL`, `FTP_COMMAND_RET_CODE`	identify the lab's machines (M4, M10)
`Attack`, `network_type`	the raw label and the network name

The starting point is **31 numeric traffic columns** (bytes, packets, durations, packet sizes, gaps between packets, throughput, retransmissions) and **8 code columns** (destination port, protocol, application protocol and its sub-code, client and server TCP flags, ICMP type, DNS query type). The labels (`attack_family`, `attack_label`, `is_attack`) and `count`, `source_dataset` are kept aside, never as features.

3 Creation / extraction

Table 2: New features, computed row by row.

Feature	Formula	Why (phase 4 finding)
`in_bytes_per_pkt`	`IN_BYTES` / `IN_PKTS`	size of the packets sent (probe vs transfer)
`out_bytes_per_pkt`	`OUT_BYTES` / `OUT_PKTS` (0 if no answer)	size of the packets received
`out_in_ratio`	`OUT_BYTES` / `IN_BYTES`	one-way traffic vs a conversation
`bytes_per_sec`	bytes (both ways) / duration	speed: floods are fast, slow attacks slow
`pkts_per_sec`	packets (both ways) / duration	packet rate
`no_answer`	`OUT_PKTS` $= 0$	closed ports, flooded victims
`one_packet`	`IN_PKTS` $= 1$	single probe or query
`syn_only`	TCP and client flags = SYN only	half-open scan
`hit_timeout`	duration ≥ 120 s	long or slow connection cut by the exporter

Phase 4 showed that the raw volumes point in opposite directions in different networks. Ratios and rates describe the *shape* of a flow rather than its size. The duration is taken as (ms + 1)/1000 seconds so that flows of 0 ms keep a finite rate.

Table 3: Created features per family (median for the numeric ones, % of behaviours for the flags; all data).

Family	B/pkt in	B/pkt out	out/in	pkt/s	no answer	one pkt	SYN only	timeout
Benign	91	163	1.27	9.7	8.4	20.0	5.0	1.2
Denial of service	99	229	1.51	2.4	**24.0**	1.7	19.8	0.6
Reconnaissance	44	40	0.91	90.9	7.6	**92.1**	**91.5**	0.0
Credential attacks	80	184	**4.11**	25.4	1.8	3.9	1.9	0.0
Web and injection	140	297	0.82	35.7	0.2	0.3	0.1	0.0
Malware and botnets	48	40	0.83	7.9	1.0	1.0	**68.2**	0.0
Exploitation and intrusion	78	91	1.28	25.0	5.6	11.6	4.2	0.7
Interception and spoofing	70	234	1.81	**100.0**	1.1	18.7	0.0	0.0
Fuzzing and crypto	78	43	**0.49**	26.5	8.6	3.7	2.4	0.0
Data theft	113	202	2.03	13.2	0.2	1.1	0.9	0.9

Interpretation. The new features capture behaviours that the raw columns only show indirectly.

- **Reconnaissance:** `one_packet` = 92% and `syn_only` = 92%. The scan is a single SYN probe, now visible in one column.
- **Denial of service:** the highest `no_answer` (24%), because flooded victims stop answering.
- **Malware and botnets:** `syn_only` = 68%; the bots open connections that are never completed.
- **Credential attacks:** the highest `out_in_ratio` (4.1); the server answers each login attempt with much more data than the request.
- **Interception and spoofing:** the fastest traffic (100 packets/s), typical of DNS spoofing.
- **Fuzzing:** the lowest `out_in_ratio` (0.49); it sends about twice as much as it receives.

Each family has a distinct value in at least one created feature, so these columns add information about the family.

4 Encoding

The code columns are **identifiers, not quantities** (decision M1): port 443 is not "more" than port 80, and a TCP flag value of 27 means "SYN + ACK + PSH + FIN", not the number 27. They are turned into yes/no (0/1) columns.

Table 4: Encoding of the code columns.

Column	Encoding	Result
Destination port	3 official ranges, one-hot	`port_well_known` (0–1023), `port_registered` (1024–49151), `port_dynamic` (49152+)
Protocol	one-hot, rare ones grouped	`proto_tcp`, `proto_udp`, `proto_other`
Client / server TCP flags	one column per bit	`client_syn`, `client_ack`, ..., `server_rst`, ... (8 flags × 2 sides)
Application protocol	one-hot of protocols with ≥ 1% of behaviours, rest grouped	9 protocols (HTTP 29.6%, TLS 17.8%, nDPI 88 16.9%, Unknown 12.0%, DNS 8.1%, NetBIOS 3.4%, codes 36, 37, 41) + `l7_other` (6.0%)

Interpretation. The 8 code columns become **32 yes/no columns**. Example: client flags 27 becomes FIN = 1, SYN = 1, PSH = 1, ACK = 1, so a model can use "the client sent a SYN" directly. The port is grouped into ranges because its number has no meaning, only its type of service; phase 4 showed that the range is strongly linked to the family. Three code columns are **not encoded**:

- `L7_SUB` is set for only 0.07% of the behaviours;
- `ICMP_TYPE` concerns 0.98% (ICMP flows), already identified by `proto_other`;
- `DNS_QUERY_TYPE` only details DNS queries (8%), already identified by `l7_dns`.

5 Transformation and scaling

5.1 Log transformation

Phase 4 showed that the volume and time columns are extremely right-skewed. $\log(1+x)$ compresses large values, keeps 0 at 0 and keeps the order of the values. It is applied to the 31 numeric columns and the 5 created ratios and rates (36 columns, all ≥ 0); the 0/1 columns are not transformed.

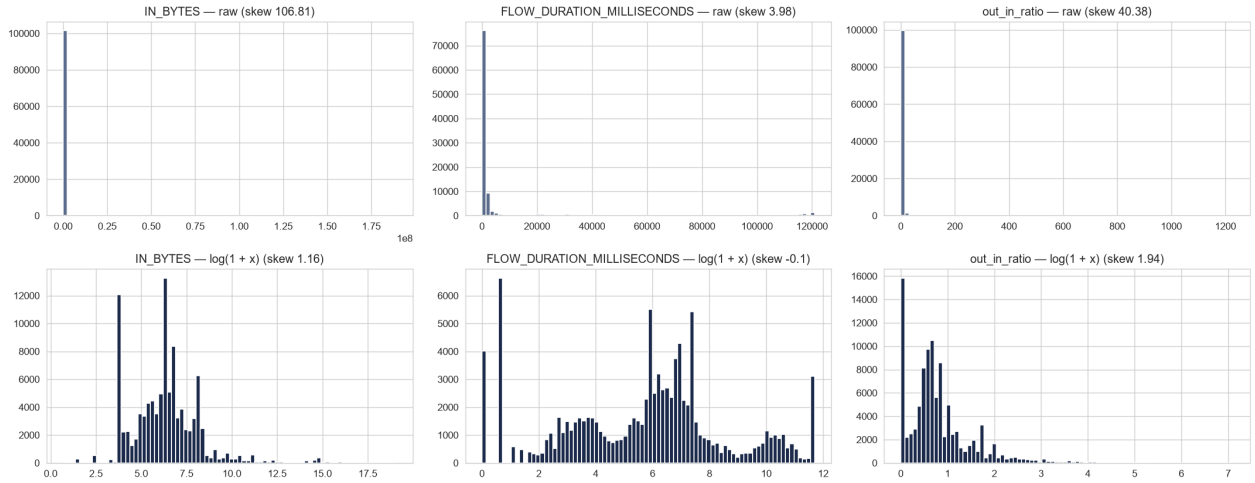


Figure 1: Three columns before (top) and after (bottom) $\log(1 + x)$ (sample).

Table 5: Skewness before and after the log transformation (examples; sample).

Column	Before	After
`IN_BYTES`	106.8	1.16
`OUT_BYTES`	100.8	−0.41
`FLOW_DURATION_MILLISECONDS`	3.98	−0.10
`LONGEST_FLOW_PKT`	39.1	−0.08
`DNS_TTL_ANSWER`	319.1	5.39
`RETRANSMITTED_OUT_PKTS`	60.9	3.56
`pkts_per_sec`	5.21	0.17
Columns with $\lvert\text{skewness}\rvert \leq 1$	0 of 36	17 of 36

Interpretation. Before the transformation, all 36 columns are highly skewed; on the raw scale almost every value sits in the first bar of the histogram. After $\log(1 + x)$, 17 columns have $\lvert\text{skewness}\rvert \leq 1$ (for example the duration: 3.98 to −0.10), and the others are strongly reduced (`IN_BYTES`: 107 to 1.2). The columns that stay above 1 are those where **most values are 0** (retransmissions, packet-size bins, minimum gaps). Their skewness comes from a spike at 0, not from a long tail, and no transformation can spread a single value. The log still removes their extreme tail.

5.2 Robust scaling

After the log, the columns still have different ranges. Each numeric column is scaled as $(x - \text{median})/\text{IQR}$, the same as scikit-learn's `RobustScaler`. The median and IQR are used instead of the mean and standard deviation because **the outliers are the attacks**: with the mean and standard deviation, a few extreme floods would set the scale for everything else. The 0/1 columns are left as they are.

Special case, IQR = 0. Five columns (the 4 retransmission columns and `DNS_TTL_ANSWER`) are 0 for more than half of the behaviours, so their IQR is 0. They are divided by their **standard deviation** instead.

Interpretation. The special case was checked in practice. A first run left these five columns unscaled, as `RobustScaler` does. Their spread (0 to 14) then dominated the PCA: the first component was made almost only of retransmissions. After the fix, the scaled columns have a typical spread of about 1 (standard deviation 0.6 to 0.8 in the sample), so no column dominates because of its unit.

6 Feature selection

The 72 candidates (36 numeric, 4 created flags, 32 encoded) go through five steps; each removal has a reason.

1. **Exact copies:** a column identical to another adds nothing.
2. **Testbed fingerprints:** columns that identify the lab's machines rather than the attack (decision M10).
3. **Near-constant columns:** one value in $\geq 99.5\%$ of all behaviours.
4. **Highly correlated pairs:** |Spearman ρ| ≥ 0.95 (the same rule as the CICIoT2023 part); the column with the lower mutual information with the family is removed.
5. **Mutual information ranking** of the remaining features.

Table 6: The 13 removed features.

Feature	Step	Reason
`TCP_WIN_MAX_IN`, `TCP_WIN_MAX_OUT`	fingerprint	window size set by each lab machine's OS (M10)
`client_urg`	near-constant	one value in 99.96% of behaviours
`server_urg`	near-constant	one value in 100.00%
`server_cwr`	near-constant	one value in 99.79%
`client_ece`	correlated	$\rho = 1.00$ with `client_cwr`
`RETRANSMITTED_OUT_PKTS`	correlated	$\rho = 0.997$ with `RETRANSMITTED_OUT_BYTES`
`RETRANSMITTED_IN_PKTS`	correlated	$\rho = 0.984$ with `RETRANSMITTED_IN_BYTES`
`proto_tcp`	correlated	$\rho = -0.976$ with `proto_udp`
`l7_dns`	correlated	$\rho = 0.971$ with `DNS_TTL_ANSWER`
`DST_TO_SRC_IAT_AVG`	correlated	$\rho = 0.965$ with `DST_TO_SRC_IAT_MAX`
`SRC_TO_DST_IAT_AVG`	correlated	$\rho = 0.958$ with `SRC_TO_DST_IAT_MAX`
`bytes_per_sec`	correlated	$\rho = 0.950$ with `SRC_TO_DST_AVG_THROUGHPUT`

Interpretation. Of the 72 candidates, **13 are removed** and **59 are kept**.

- **No exact copies:** phase 3 had already dropped the copy columns (M7, M8).
- **2 fingerprints:** phase 4 showed that each family uses one TCP window size, set by the operating system of the lab machine. A model would learn the testbed, not the attack.
- **3 near-constant flags** carry almost no information.
- **8 correlated pairs:** retransmitted packets duplicate retransmitted bytes; `proto_tcp` is almost the opposite of `proto_udp`; `l7_dns` duplicates `DNS_TTL_ANSWER` (only DNS flows have an answer TTL); the average gap duplicates the largest gap; `bytes_per_sec` duplicates the existing throughput.

Only one of the 9 created features is removed (a near-copy of an existing column), so the other 8 add new information.

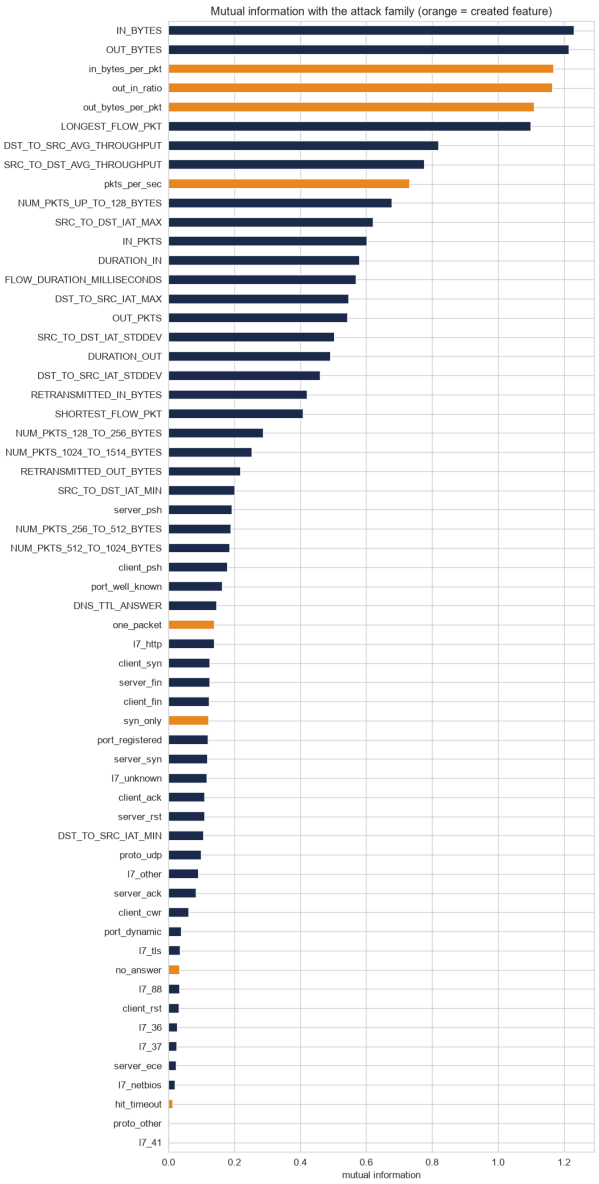


Figure 2: Mutual information of the 59 final features with the attack family (orange = created feature; sample).

Interpretation. The most informative features are the volume and packet-size features: `IN_BYTES` (1.23), `OUT_BYTES` (1.22), then three **created** features, `in_bytes_per_pkt` (1.17), `out_in_ratio` (1.17) and `out_bytes_per_pkt` (1.11). **Three of the top five are created features**, which justifies their creation. The 0/1 columns score lower (at most 0.19). This is expected: a yes/no column can carry at most $\ln 2 \approx 0.69$ of information. It does not make them useless (`syn_only` alone marks 92% of the scans), so they are compared with each other, not with the numeric columns.

7 Final feature set

7.1 From 53 raw columns to 59 features

The four NetFlow v3 datasets share **53 traffic columns** (plus the two label columns). The table follows every column from the raw files to the final feature set.

Table 7: How the number of columns changes, stage by stage.

Stage	Columns	What changes
Raw NetFlow v3 files	53	the same 53 columns in UNSW, CIC-2018, ToN-IoT and BoT-IoT
Phases 2–3 (cleaning, merge)	46	−2 IP addresses (M3), −2 bytes-per-second columns with NaN / inf (U2, B2, C3, T3), −2 IP packet-length copies (M7), −1 `TCP_FLAGS` and −1 `ICMP_IPV4_TYPE` copies (M8); +1 because `L7_PROTO` is split into `L7_MAIN` and `L7_SUB` (U3)
Phase 5, step 0 (never features)	39	−7: 2 timestamps (M2), source port and DNS query ID (M5), 2 TTL (M4), FTP return code (M10). Left: **31 numeric** and **8 code** columns
Phase 5, encoding	63	the 8 code columns become **32** yes/no columns; `L7_SUB`, `ICMP_TYPE` and `DNS_QUERY_TYPE` are not encoded (too rare or already covered)
Phase 5, creation	72	+9 created features (5 ratios and rates, 4 behaviour flags): the **72 candidates**
Phase 5, selection	**59**	−13: 2 fingerprints, 3 near-constant, 8 correlated (section 6)

Table 8: Candidates, removed and kept, per group.

Group	Candidates	Removed	Kept	Removed columns
Original numeric	31	6	**25**	`TCP_WIN_MAX_IN`, `TCP_WIN_MAX_OUT` (fingerprints); `RETRANSMITTED_IN_PKTS`, `RETRANSMITTED_OUT_PKTS`, `SRC_TO_DST_IAT_AVG`, `DST_TO_SRC_IAT_AVG` (correlated)
Created	9	1	**8**	`bytes_per_sec` (correlated with the existing throughput)
Encoded	32	6	**26**	`client_urg`, `server_urg`, `server_cwr` (near-constant); `client_ece`, `proto_tcp`, `l7_dns` (correlated)
Total	**72**	**13**	**59**	

Table 9: The 59 final features by what they describe.

What it describes	Count	Features
Volume	6	`IN_BYTES`, `OUT_BYTES`, `IN_PKTS`, `OUT_PKTS`, `RETRANSMITTED_IN_BYTES`, `RETRANSMITTED_OUT_BYTES`
Packet sizes	7	`LONGEST_FLOW_PKT`, `SHORTEST_FLOW_PKT`, and the 5 size bins `NUM_PKTS_UP_TO_128_BYTES` ... `NUM_PKTS_1024_TO_1514_BYTES`
Time and rate	11	`FLOW_DURATION_MILLISECONDS`, `DURATION_IN`, `DURATION_OUT`, the 2 average throughputs, and the minimum, maximum and standard deviation of the gaps between packets in each direction (6)
DNS	1	`DNS_TTL_ANSWER`
Original numeric columns: 25		
Created: ratios and rates	4	`in_bytes_per_pkt`, `out_bytes_per_pkt`, `out_in_ratio`, `pkts_per_sec`
Created: behaviour flags	4	`no_answer`, `one_packet`, `syn_only`, `hit_timeout`
Created features: 8		
Destination port range	3	`port_well_known`, `port_registered`, `port_dynamic`
Protocol	2	`proto_udp`, `proto_other` (TCP = both 0)
Client TCP flags	6	`client_fin`, `client_syn`, `client_rst`, `client_psh`, `client_ack`, `client_cwr`
Server TCP flags	6	`server_fin`, `server_syn`, `server_rst`, `server_psh`, `server_ack`, `server_ece`
Application protocol	9	`l7_http`, `l7_tls`, `l7_unknown`, `l7_netbios`, `l7_88`, `l7_36`, `l7_37`, `l7_41`, `l7_other` (DNS flows are recognised by `DNS_TTL_ANSWER`)
Encoded columns: 26		
Total	**59**	

Interpretation.

- **Most features are original NetFlow measurements** (25 of 59). Only 6 of the 31 numeric traffic columns were removed: 2 because they identify the lab's machines, 4 because they repeated another column.
- **8 of the 9 created features are kept;** the only one removed, `bytes_per_sec`, duplicated an existing throughput column.
- **The 26 encoded columns turn 5 code columns into usable yes/no information:**
  - port range, protocol and application protocol: the type of service;
  - 12 TCP flags: how the connection behaved.
- **Two encoded groups lost one column with no loss of information:**
  - `proto_tcp` is implied when `proto_udp` and `proto_other` are both 0;
  - `l7_dns` is carried by `DNS_TTL_ANSWER`.

7.2 Justification of the final set

Interpretation. The final set has **59 features**: 25 original numeric columns, 8 created features and 26 encoded columns.

- **Based on evidence:** every feature comes from a phase 2 decision or a phase 4 finding.

- **No identifier of the network or the moment:** no IP address, timestamp or TTL, and the TCP window is removed as a fingerprint.
- **No duplicates:** no pair has $|\rho| \geq 0.95$.
- **No useless columns:** no column is near-constant.
- **Ready for a model:** every numeric column is log-transformed and robust-scaled, and every code is one-hot encoded.

The final dataset keeps all 17,551,848 behaviours with the 59 features and 6 label/weight columns (634 MB):

- data: `data/processed/netflow_features.parquet`;
- data dictionary: `tables/NetFlow_fe_dictionary.csv`.

Table 10: The 59 final features, ordered by mutual information (MI) with the family. Numeric features are log-transformed and robust-scaled; 0/1 features are not transformed.

Feature	Meaning	Origin	MI
IN_BYTES	bytes sent by the client	original	1.231
OUT_BYTES	bytes sent back by the server	original	1.216
in_bytes_per_pkt	average size of the packets sent	created	1.169
out_in_ratio	bytes received per byte sent	created	1.165
out_bytes_per_pkt	average size of the packets received	created	1.110
LONGEST_FLOW_PKT	largest packet of the flow	original	1.100
DST_TO_SRC_AVG_THROUGHPUT	average throughput server to client	original	0.820
SRC_TO_DST_AVG_THROUGHPUT	average throughput client to server	original	0.778
pkts_per_sec	packets per second (both directions)	created	0.733
NUM_PKTS_UP_TO_128_BYTES	packets of at most 128 bytes	original	0.679
SRC_TO_DST_IAT_MAX	largest gap between client packets	original	0.622
IN_PKTS	packets sent by the client	original	0.602
DURATION_IN	duration of the client-to-server stream	original	0.579
FLOW_DURATION_MILLISECONDS	duration of the flow	original	0.570
DST_TO_SRC_IAT_MAX	largest gap between server packets	original	0.547
OUT_PKTS	packets sent back by the server	original	0.544
SRC_TO_DST_IAT_STDDEV	spread of the gaps between client packets	original	0.505
DURATION_OUT	duration of the server-to-client stream	original	0.492
DST_TO_SRC_IAT_STDDEV	spread of the gaps between server packets	original	0.462
RETRANSMITTED_IN_BYTES	bytes retransmitted by the client	original	0.420
SHORTEST_FLOW_PKT	smallest packet of the flow	original	0.409
NUM_PKTS_128_TO_256_BYTES	packets of 128–256 bytes	original	0.288
NUM_PKTS_1024_TO_1514_BYTES	packets of 1024–1514 bytes	original	0.254
RETRANSMITTED_OUT_BYTES	bytes retransmitted by the server	original	0.219
SRC_TO_DST_IAT_MIN	smallest gap between client packets	original	0.202
server_psh	1 if the server set the PSH flag	encoded	0.193
NUM_PKTS_256_TO_512_BYTES	packets of 256–512 bytes	original	0.190
NUM_PKTS_512_TO_1024_BYTES	packets of 512–1024 bytes	original	0.186
client_psh	1 if the client set the PSH flag	encoded	0.180
port_well_known	1 if the destination port is 0–1023	encoded	0.164
DNS_TTL_ANSWER	time-to-live of the DNS answer	original	0.146
one_packet	1 if the client sent a single packet	created	0.140
l7_http	1 if the application protocol is HTTP	encoded	0.139
client_syn	1 if the client set the SYN flag	encoded	0.126
server_fin	1 if the server set the FIN flag	encoded	0.125
client_fin	1 if the client set the FIN flag	encoded	0.124
syn_only	1 if TCP and the client sent only SYN (half-open scan)	created	0.123
port_registered	1 if the destination port is 1024–49151	encoded	0.120

Feature	Meaning	Origin	MI
server_syn	1 if the server set the SYN flag	encoded	0.119
l7_unknown	1 if the application protocol is unknown	encoded	0.118
client_ack	1 if the client set the ACK flag	encoded	0.111
server_rst	1 if the server set the RST flag	encoded	0.111
DST_TO_SRC_IAT_MIN	smallest gap between server packets	original	0.106
proto_udp	1 if UDP	encoded	0.100
l7_other	1 if a rare application protocol	encoded	0.091
server_ack	1 if the server set the ACK flag	encoded	0.084
client_cwr	1 if the client set the CWR flag	encoded	0.062
port_dynamic	1 if the destination port is 49152 or more	encoded	0.039
l7_tls	1 if the application protocol is TLS	encoded	0.037
no_answer	1 if the server sent nothing back	created	0.035
l7_88	1 if the application protocol is nDPI code 88	encoded	0.034
client_rst	1 if the client set the RST flag	encoded	0.033
l7_36	1 if the application protocol is nDPI code 36	encoded	0.028
l7_37	1 if the application protocol is nDPI code 37	encoded	0.026
server_ece	1 if the server set the ECE flag	encoded	0.025
l7_netbios	1 if the application protocol is NetBIOS	encoded	0.020
hit_timeout	1 if the flow lasted 120 s or more	created	0.014
proto_other	1 if another protocol (ICMP, ...)	encoded	0.004
l7_41	1 if the application protocol is nDPI code 41	encoded	0.003

8 PCA: a 2-D view of the families

Principal component analysis builds new axes (components) as combinations of the features, ordered by how much of the variation they explain. Two components give a map of the data: if the families form separate clouds, the features separate them well. PCA is fitted on the sample with the 59 final features.

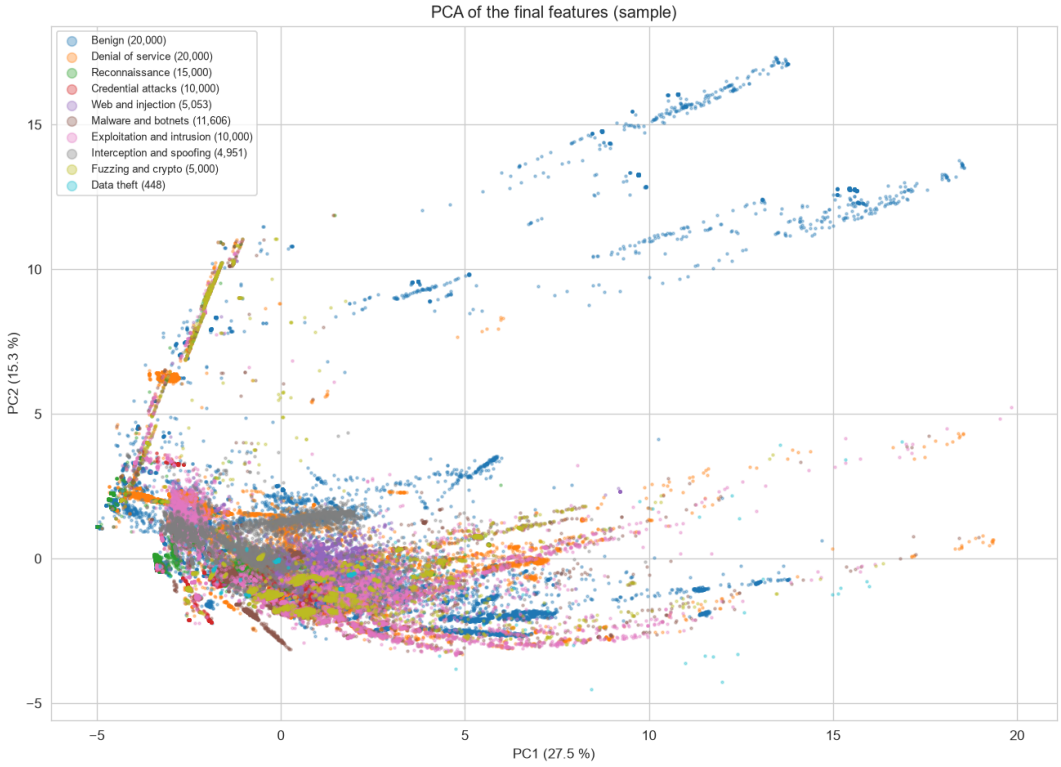


Figure 3: The sample projected on the first two principal components, coloured by family.

Interpretation.

- **Variance:** the first two components explain **42.8%** of the variation, and 10 components explain 85.8%, so the 59 features do not reduce to two dimensions.
- **The two axes:** PC1 is the **size of the flow** (number of packets and packets of each size). PC2 contrasts the smallest packet with the answer size and throughput.
- **Separate clusters:** Reconnaissance is a tight group on the left (tiny one-packet probes), and Fuzzing lies along its own line.
- **Benign** spreads over the whole map: normal traffic is the most varied.
- **Most attack families overlap** in the centre.

As in phase 4, no single direction separates the families: a model will need many features together. PCA is used here to look at the data; the features themselves are kept for interpretability.

9 Techniques not used, and why

Technique	Used?	Why
Date / hour-of-day features	no	timestamps reveal the lab's attack schedule (M2); a model would learn *when* the lab ran each attack
Polynomial / interaction terms	no	the relationships found in phase 4 are monotonic, not curved; squared terms of about 60 columns would add about 1,800 columns without a reason
Binning of bytes, packets or durations	only the port	the log transform keeps the full detail; bins would throw it away. The port is binned because its number has no meaning, only its range
Target encoding	no	it builds a feature from the label (risk of leakage); no column has so many categories that one-hot is impossible
Train/test split, model-based importance	no	the project stops before modelling; mutual information ranks the features without a model

10 Conclusion

1. **The created features earn their place:** 8 of the 9 survive the selection, and 3 are among the 5 most informative features. Ratios and behaviour flags (one packet, SYN only, no answer) capture scans, floods and conversations directly.
2. **Log transformation and robust scaling are necessary:** the data are extremely skewed, and the outliers are the attacks themselves. The special case IQR = 0 had to be handled so that no column dominates.
3. **The final set keeps only traffic behaviour:** 59 features, with no column that identifies the network, the machine or the moment. This matters for SQ3, because a model must not recognise the lab instead of the attack.

Before modelling (outside this project): split the data first, then refit the scaling and the selection on the training part only.